



HOUGHTON™

Revision Date: 07-14-2016

Version 1

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1.

Product identifier

Product Code(s):	42070795-M
Product Name	HOCUT 795 MPB
Product Registration number	
Denmark	-
Norway	-
Sweden	-
EC #	
Pure substance/preparation	Contains (Z)-Octadec-9-enylamine, ethoxylated , 1-Aminopropan-2-ol, Dodecan-1-ol, ethoxylated , 3-Iodo-2-propynyl butylcarbamate

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended use	Metalworking fluid
Uses advised against	Any other purpose.

1.3. Details of the supplier of the safety data sheet

Manufacturer, Importer, Supplier

Houghton Singapore 22 Boon Lay Way, #01-61 Trade Hub 21, Singapore 609968 Tel: +65-6235-1544	Thai Houghton 1993 Co., Ltd. 77/105-106, 25th Floor Sinn Sathorn Tower Krungdhonburi Road Klongsarn, Bangkok 10600, Thailand Tel: + 66-2-440-1262	Houghton Australia Pty. Ltd. 287 Wickham Road Moorabbin, Victoria Australia, 3189 +61-3-9555-0344
D A Stuart India Private Ltd. (a Houghton Company) 502&503, Montreal Business Centre 5th Floor, Baner, Pune 411 045 Maharashtra, India Tel: +91-20-6724-5300		

1.4. Emergency telephone number

3E Company: (+)1 760 476 3961 (Code 333938)

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

Regulation (EC) No 1272/2008

Skin corrosion/irritation	Category 2 - (H315)
Serious eye damage/eye irritation	Category 1 - (H318)
Chronic aquatic toxicity	Category 2 - (H411)

Contains 3-Iodo-2-propynyl butylcarbamate. May produce an allergic reaction.

2.2. Label Elements

Contains (Z)-Octadec-9-enylamine, ethoxylated , 1-Aminopropan-2-ol, Dodecan-1-ol, ethoxylated , 3-Iodo-2-propynyl butylcarbamate



Signal Word
DANGER

Hazard Statements

H315 - Causes skin irritation

H318 - Causes serious eye damage

H411 - Toxic to aquatic life with long lasting effects

EUH208 - Contains 3-Iodo-2-propynyl butylcarbamate. May produce an allergic reaction.

Precautionary Statements - EU (§28, 1272/2008)

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water

P332 + P313 - If skin irritation occurs: Get medical advice/attention

P362 + P364 - Take off contaminated clothing and wash it before reuse

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor/physician

2.3. Other hazards

No information available

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.1. Substances / 3.2. Mixtures

This product is a mixture. Health hazard information is based on its ingredients

Chemical Name	EC-No	CAS-No	Weight %	Classification (Reg. 1272/2008)	REACH Registration Number
Highly refined, low viscosity mineral oils/hydrocarbons (Viscosity >7 - <20.5 cSt @40°C)	-	-	10% - 25%	Asp. Tox. 1 (H304) (EUH066)	-
Highly refined base oil (Viscosity >20.5 cSt @40°C)	-	-	2.5% - 10%	**	-
Neutralised Dicyclohexylamine	202-980-7	101-83-7*	2.5% - 10%	Acute Tox. 4 (H302) Aquatic Acute 1 (H400) Aquatic Chronic 1 (H410)	01-2119493354-33-xxx x

Alcohol, C10-12, ethoxylated, propoxylated	-	68154-97-2	2.5% - 10%	Skin Irrit. 2 (H315) Eye Irrit. 2 (H319) STOT SE 3 (H335)	no data available
(Z)-Octadec-9-enylamine, ethoxylated	-	26635-93-8	2.5% - 10%	Acute Tox. 4 (H302) Eye Dam. 1 (H318) Skin Irrit. 2 (H315) Aquatic Acute 1 (H400) Aquatic Chronic 1 (H410)	no data available
Neutralised 1-Aminopropan-2-ol	201-162-7	78-96-6*	1% - 2.5%	Acute Tox. 4 (H312)	01-2119475331-43-xxx x
1-Aminopropan-2-ol	201-162-7	78-96-6	1% - 2.5%	Skin Corr. 1B (H314) Acute Tox. 4 (H312)	01-2119475331-43-xxx x
Neutralised nonanoic acid	203-931-2	112-05-0*	1% - 2.5%	No data available	01-2119529247-37-xxx x
2,2',2"-Nitrilotriethanol	203-049-8	102-71-6	1% - 2.5%	**	01-2119486482-31-xxx x
Neutralised boric acid	233-139-2	10043-35-3*	1% - 2.5%	Repr. 1B (H360FD)	01-2119486683-25-xxx x
Dodecan-1-ol, ethoxylated	-	9002-92-0	0% - 1%	Eye Dam. 1 (H318) Aquatic Acute 1 (H400)	no data available
3-Iodo-2-propynyl butylcarbamate	259-627-5	55406-53-6	0% - 1%	Acute Tox. 3 (H331) Acute Tox. 4 (H302) STOT RE 1 (H372) Eye Dam. 1 (H318) STOT SE 3 (H335) Skin Sens. 1 (H317) Aquatic Acute 1 (H400) Aquatic Chronic 1 (H410)	no data available

Additional information

SVHC ...Contains... BORIC ACID Concentration: < 5,5% See Section 11 for more information See Section 15 for additional information on base oils.

** Substances for which there are Community workplace exposure limits

Full text of H- and EUH-phrases: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first-aid measures

General advice	Do not get in eyes, on skin, or on clothing. Immediate medical attention is required. May produce an allergic reaction.
Inhalation	Move to fresh air.
Skin contact	Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing before re-use. If symptoms persist, call a physician. May cause an allergic skin reaction.
Eye contact	Keep eye wide open while rinsing. Immediately flush with plenty of water. After initial flushing, remove any contact lenses and continue flushing for at least 15 minutes. Do not rub affected area. Seek immediate medical attention/advice.
Ingestion	Clean mouth with water. Drink plenty of water. Do not induce vomiting without medical advice.
Protection of First-aiders	Use personal protective equipment. Avoid contact with skin, eyes and clothing.

4.2. Most important symptoms and effects, both acute and delayed

Main Symptoms Redness, Itching, Rash, Eye damage/irritation, May cause allergic skin reaction

4.3. Indication of immediate medical attention and special treatment needed

Notes to physician May cause sensitization of susceptible persons. Treat symptomatically.

SECTION 5: FIRE FIGHTING MEASURES**5.1. Extinguishing media****Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment; Use CO2, dry chemical, or foam, Water spray or fog

Extinguishing media which shall not be used for safety reasons

None

5.2. Special hazards arising from the substance or mixture**Special Hazard**

Water runoff can cause environmental damage.

Hazardous Decomposition Products

None under normal use

5.3. Advice for firefighters**Special protective equipment for fire-fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear

SECTION 6: ACCIDENTAL RELEASE MEASURES**6.1. Personal precautions, protective equipment and emergency procedures**

Ensure adequate ventilation. Use personal protective equipment. Avoid contact with skin, eyes and clothing.

Advice for non-emergency personnel

Evacuate personnel to safe areas. Material can create slippery conditions.

Advice for emergency responders For personal protection see section 8.

6.2. Environmental precautions

Prevent further leakage or spillage if safe to do so. Do not flush into surface water or sanitary sewer system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

6.3. Methods and materials for containment and cleaning up

Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Dike to collect large liquid spills.

6.4. Reference to other sections

See Section 8/12/13 for additional information

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Wear personal protective equipment. Avoid contact with skin, eyes and clothing. Remove and wash contaminated clothing before re-use. Ensure adequate ventilation. Do not eat, drink or smoke when using this product. Handle in accordance with good industrial hygiene and safety practice.

7.2. Conditions for safe storage, including any incompatibilities

Technical measures/Storage conditions

Keep container tightly closed in a dry and well-ventilated place.

Recommended Shelf Life

Shelf life 12 months.

Incompatible Materials

Strong oxidizing agents, Strong acids, Strong bases

7.3. Specific end uses

Specific use(s)

Metalworking fluid

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Chemical Name	European Union	United Kingdom	France	Spain
Highly refined, low viscosity mineral oils/hydrocarbons (Viscosity >7 - <20.5 cSt @40°C)				VLA-EC: 10 mg/m ³ VLA-ED: 5 mg/m ³
Highly refined base oil (Viscosity >20.5 cSt @40°C)				VLA-EC: 10 mg/m ³ VLA-ED: 5 mg/m ³
2,2',2''-Nitrilotriethanol				TWA: 5 mg/m ³

Chemical Name	Germany	Italy	Portugal	The Netherlands
Highly refined, low viscosity mineral oils/hydrocarbons (Viscosity >7 - <20.5 cSt @40°C)		TWA: 5 mg/m ³	TWA: 5 mg/m ³ STEL: 10 mg/m ³	TWA: 5 mg/m ³
Highly refined base oil (Viscosity >20.5 cSt @40°C)		TWA: 5 mg/m ³	TWA: 5 mg/m ³ STEL: 10 mg/m ³	TWA: 5 mg/m ³
1-Aminopropan-2-ol	TWA: 2 ppm TWA: 5.8 mg/m ³			
2,2',2''-Nitrilotriethanol	MAK: 5 mg/m ³ Ceiling / Peak: 20 mg/m ³		TWA: 5 mg/m ³	

Chemical Name	Austria	Switzerland	Poland	Ireland
Highly refined, low viscosity mineral oils/hydrocarbons			TWA: 5 mg/m ³ STEL: 10 mg/m ³	STEL: 10 mg/m ³ TWA: 5 mg/m ³

(Viscosity >7 - <20.5 cSt @40°C)				(Mist)
Highly refined base oil (Viscosity >20.5 cSt @40°C)			TWA: 5 mg/m ³ STEL: 10 mg/m ³	STEL: 10 mg/m ³ TWA: 5 mg/m ³ (Mist)
2,2',2''-Nitrilotriethanol	STEL 1.6 ppm MAK: 0.8 ppm MAK: 5 mg/m ³			TWA: 5 mg/m ³

Chemical Name	Finland	Denmark	Norway	Sweden
Highly refined, low viscosity mineral oils/hydrocarbons (Viscosity >7 - <20.5 cSt @40°C)	TWA: 5mg/m ³ (Öljysumu)	TWA: 1 mg/m ³ (Olietåge)	TWA: 1 mg/m ³ (Oljetåke)	LLV: 1 mg/m ³ STV: 3 mg/m ³ (Oljedimma)
Highly refined base oil (Viscosity >20.5 cSt @40°C)	TWA: 5mg/m ³ (Öljysumu)	TWA: 1 mg/m ³ (Olietåge)	TWA: 1 mg/m ³ (Oljetåke)	LLV: 1 mg/m ³ STV: 3 mg/m ³ (Oljedimma)
2,2',2''-Nitrilotriethanol	TWA: 5 ppm	TWA: 0.5 ppm TWA: 3.1 mg/m ³	TWA: 5 mg/m ³ STEL: 10 mg/m ³	LLV: 5 mg/m ³ STV: 10 mg/m ³

Chemical Name	Czech Republic	Hungary	Bulgaria	Romania
Highly refined, low viscosity mineral oils/hydrocarbons (Viscosity >7 - <20.5 cSt @40°C)	TWA: 5 mg/m ³ Ceiling: 10 mg/m ³	TWA: 5 mg/m ³	TWA: 5 mg/m ³	TWA: 5 mg/m ³ STEL: 10 mg/m ³
Highly refined base oil (Viscosity >20.5 cSt @40°C)	TWA: 5 mg/m ³ Ceiling: 10 mg/m ³	TWA: 5 mg/m ³	TWA: 5 mg/m ³	TWA: 5 mg/m ³ STEL: 10 mg/m ³
2,2',2''-Nitrilotriethanol	Ceiling: 10 mg/m ³ TWA: 5 mg/m ³		TWA: 3.0 mg/m ³	

Chemical Name	Greece	Cyprus	Turkey	Malta
Highly refined, low viscosity mineral oils/hydrocarbons (Viscosity >7 - <20.5 cSt @40°C)	TWA: 5 mg/m ³			
Highly refined base oil (Viscosity >20.5 cSt @40°C)	TWA: 5 mg/m ³			

Chemical Name	Belgium	Luxembourg	Iceland	Croatia
Highly refined, low viscosity mineral oils/hydrocarbons (Viscosity >7 - <20.5 cSt @40°C)	TWA: 5 mg/m ³ STEL: 10 mg/m ³			
Highly refined base oil (Viscosity >20.5 cSt @40°C)	TWA: 5 mg/m ³ STEL: 10 mg/m ³			
2,2',2''-Nitrilotriethanol	TWA: 5 mg/m ³		TWA: 5 mg/m ³	

Chemical Name	Russia	Estonia	Latvia	Lithuania
Highly refined, low viscosity mineral oils/hydrocarbons (Viscosity >7 - <20.5 cSt @40°C)			TWA: 5 mg/m ³	TWA: 1 mg/m ³ STEL: 3 mg/m ³
Highly refined base oil (Viscosity >20.5 cSt @40°C)			TWA: 5 mg/m ³	TWA: 1 mg/m ³ STEL: 3 mg/m ³
1-Aminopropan-2-ol	MAC: 1 mg/m ³ Skin			
2,2',2''-Nitrilotriethanol	MAC: 3 mg/m ³	STEL: 10 mg/m ³ TWA: 5 mg/m ³		TWA: 5 mg/m ³ STEL: 10 mg/m ³

Chemical Name	Belarus	Ukraine	Slovakia	Slovenia
Highly refined, low viscosity mineral oils/hydrocarbons (Viscosity >7 - <20.5 cSt @40°C)			TWA: 5mg/m ³	
Highly refined base oil (Viscosity >20.5 cSt @40°C)			TWA: 5mg/m ³	
2,2',2''-Nitrilotriethanol				TWA: 5 mg/m ³

Chemical Name	Serbia	Macedonia	Liechtenstein	South Africa
2,2',2''-Nitrilotriethanol		TWA: 5 mg/m ³		

Legend:

(s) - Skin

TWA - Time-Weighted Average

STEL - Short Term Exposure Limit

Ceiling - Ceiling Value

Hydrocarbon solvent vapor mixtures which do not have substance specific occupational exposure limits may be evaluated by the Reciprocal Calculation Procedure (RCP) which assigns a recommended occupational exposure limit based on the mass composition and hydrocarbon group guidance values (GGVs). Applicable recommended occupational exposure limits are shown in the table below.

Chemical Name	RCP OEL	Manufacturer
Paraffin oils 8012-95-1		TWA: 5 mg/m ³ excluding metal working fluids, highly & severely refined

Chemical Name	RCP OEL	Manufacturer
Distillates (petroleum), hydrotreated middle 64742-46-7	RCP: TWA 1200 mg/m ³ 143ppm	

Derived No Effect Level (DNEL)**Workers Systemic toxicity**

Chemical Name	Long term - Oral exposure	Long term - Dermal exposure	Long term - Inhalation exposure	Short term - Oral Exposure	Short term - Dermal exposure	Short term - Inhalation exposure
Neutralised 1-Aminopropan-2-ol		8.5 mg/kg				
1-Aminopropan-2-ol		8.5 mg/kg				
Neutralised nonanoic acid		8.33 mg/kg	58.8 mg/m ³		50 mg/kg	117 mg/m ³
Neutralised boric acid		392 mg/kg	8.3 mg/m ³			

Workers Local effects**Consumers Systemic toxicity**

Chemical Name	Long term - Oral exposure	Long term - Dermal exposure	Long term - Inhalation exposure	Short term - Oral Exposure	Short term - Dermal exposure	Short term - Inhalation exposure
Neutralised 1-Aminopropan-2-ol		2.1 mg/kg	0.67 mg/m ³			
1-Aminopropan-2-ol		2.1 mg/kg	0.67 mg/m ³			
Neutralised nonanoic acid	4.17 mg/kg	4.17 mg/kg	14.5 mg/m ³	25 mg/kg	25 mg/kg	29 mg/m ³
Neutralised boric acid	0.98 mg/kg	196 mg/kg	4.15 mg/m ³	0.98 mg/kg		

Consumers Local effects**Predicted No Effect Concentration (PNEC)**

Chemical Name	Fresh water	Sea water	Fresh water sediment	Sea sediment	Soil
Neutralised 1-Aminopropan-2-ol	0.0327 mg/l	0.00327 mg/l	0.177 mg/kg	0.0177 mg/kg	0.0161 mg/kg
1-Aminopropan-2-ol	0.0327 mg/l	0.00327 mg/l	0.177 mg/kg	0.0177 mg/kg	0.0161 mg/kg
Neutralised nonanoic acid	0.36 mg/L	0.036 mg/L	8.5 mg/kg	0.85 mg/kg	1.48 mg/kg
Neutralised boric acid	1.35 mg/l	1.35 mg/l	1.8 mg/kg	1.8 mg/kg	5.4 mg/kg
3-Iodo-2-propynyl butylcarbamate	0.0005 mg/L				0.005 mg/kg

8.2. Exposure controls

Engineering Measures	Ensure adequate ventilation, especially in confined areas. Eyewash stations.
Personal protective equipment	
Eye Protection	Eye protection must conform to standard EN 166. Tightly fitting safety goggles.
Hand Protection	Protective gloves complying with EN 374. Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. Also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.
Skin and body protection	Long sleeved clothing.
Respiratory protection	No special protective equipment required. In case of mist, spray or aerosol exposure wear suitable personal respiratory protection and protective suit.
Hygiene measures	Wear personal protective equipment. Avoid contact with skin, eyes and clothing. Remove and wash contaminated clothing before re-use. Do not eat, drink or smoke when using this product. Handle in accordance with good industrial hygiene and safety practice. Regular cleaning of equipment, work area and clothing is recommended.
Environmental Exposure Controls	Do not allow material to contaminate ground water system.
Thermal hazards	None under normal use conditions

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES**9.1. Information on basic physical and chemical properties**

Physical state @20°C	liquid	Appearance	clear blue
Odor	No information available	Odor Threshold	Not Applicable
<u>Property</u>	<u>Values</u>		<u>Note</u>
pH	9.6		@5%
Melting Point / Freezing Point	No information available		
Boiling point/boiling range	No information available		
Flash point	Not Applicable		
Evaporation rate	No information available		
Flammability (solid, gas)	No information available		
Flammability Limits in Air			
upper flammability limit	No information available		
Lower flammability limit	No information available		
Vapor pressure	No information available		
Vapor density	No information available		
Relative density	0.966		g/cm3 @20°C
Solubility(ies)	Emulsifies		
Partition coefficient: n-octanol/water	Not Applicable		
Autoignition temperature	No information available		
Decomposition temperature	No information available		
Viscosity, kinematic	> 30 cSt @ 40 °C		ASTM D 445
Explosive properties	Not Applicable		
Oxidizing Properties	Not Applicable		

9.2. Other information

Viscosity, kinematic (100°C)	No information available
Pour point	No information available
VOC Content (ASTM E-1868-10)	No information available
VOC content	No information available

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

None under normal use conditions

10.2. Chemical stability

Stable under normal conditions

10.3. Possibility of hazardous reactions

None under normal use conditions

10.4. Conditions to avoid

Extremes of temperature and direct sunlight

10.5. Incompatible Materials

Strong oxidizing agents, Strong acids, Strong bases

10.6. Hazardous decomposition products

None under normal use conditions

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on toxicological effects

Product Information - Principle Routes of Exposure

Inhalation	None known
Eye contact	May result in permanent damage including blindness
Skin contact	Irritating to skin; Repeated or prolonged skin contact may cause allergic reactions with susceptible persons
Ingestion	None known

Acute toxicity - Product Information

Product does not present an acute toxicity hazard based on known information.

Acute toxicity - Component Information

Chemical Name	LD50 Oral (Rat)	LD50 Dermal (Rat/Rabbit)	LC50 Inhalation
Highly refined, low viscosity mineral oils/hydrocarbons (Viscosity >7 - <20.5 cSt @40°C)	>2000 mg/kg	>2000 mg/kg	
Highly refined base oil (Viscosity >20.5 cSt @40°C)	>2000 mg/kg	>2000 mg/kg	
Alcohol, C10-12, ethoxylated, propoxylated	>2000 mg/kg (Rat)	>2000 mg/kg (Rabbit)	
Neutralised 1-Aminopropan-2-ol	2098 mg/kg (Rat)	1851 mg/kg (Rabbit)	
1-Aminopropan-2-ol	2098 mg/kg (Rat)	1851 mg/kg (Rabbit)	

Neutralised nonanoic acid	9000 mg/kg (Rat)	9000 mg/kg (Rabbit)	
2,2',2''-Nitrilotriethanol		> 16 mL/kg (Rat) > 2000 mg/kg (Rabbit)	
Neutralised boric acid	3500 mg/kg (Rat)	> 2000 mg/kg (Rabbit)	> 0.16 mg/L (Rat) 4 h
Dodecan-1-ol, ethoxylated	8600 mg/kg (Rat)		
3-Iodo-2-propynyl butylcarbamate	1470 mg/kg (Rat)	>2000 mg/kg (Rabbit)	0.67 mg/L (Rat) 4 h

Skin corrosion/irritation Irritating to skin.

Serious eye damage/eye irritation Causes severe eye damage.

Sensitization

Respiratory Sensitization

Based on available data, the classification criteria are not met.

Skin sensitization

May cause an allergic skin reaction.

Germ Cell Mutagenicity

Based on available data, the classification criteria are not met.

Carcinogenicity

Based on available data, the classification criteria are not met.

Reproductive toxicity

Contains a known or suspected reproductive toxin.

Specific target organ systemic toxicity (single exposure)

Based on available data, the classification criteria are not met

Specific target organ systemic toxicity (repeated exposure)

Based on available data, the classification criteria are not met

Aspiration hazard

Based on available data, the classification criteria are not met.

Other adverse effects

The document number 031 published 05-2013 by the DGUV Germany has been considered in the production of this safety data sheet (SDS). As stated in this document the classification of dicyclohexylamine is currently inconsistent between the manufacturer and the REACH classification. The classification used in this document is based on the European harmonised classification.

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

Chemical Name	Toxicity to algae	Toxicity to fish	Toxicity to microorganisms	Toxicity to daphnia and other aquatic invertebrates
Neutralised Dicyclohexylamine		62: 96 h Brachydanio rerio mg/L LC50 static		
(Z)-Octadec-9-enylamine, ethoxylated				0.1 - 1: 48 h Daphnia magna mg/L EC50
Neutralised 1-Aminopropan-2-ol	32.7: 72 h Scenedesmus subspicatus mg/L EC50	2520: 96 h Pimephales promelas mg/L LC50 210: 96 h Carassius auratus mg/L LC50 215 - 464: 96 h Leuciscus idus mg/L LC50		108.82: 48 h Daphnia magna mg/L EC50
1-Aminopropan-2-ol	32.7: 72 h Scenedesmus subspicatus mg/L EC50	2520: 96 h Pimephales promelas mg/L LC50 210: 96 h Carassius auratus		108.82: 48 h Daphnia magna mg/L EC50

		mg/L LC50 215 - 464: 96 h <i>Leuciscus idus</i> mg/L LC50		
Neutralised nonanoic acid	60: 72 h <i>Pseudokirchneriella subcapitata</i> mg/L EC50	93.4-115: 96 h <i>Pimephales promelas</i> mg/L LC50 flow-through 68-121: 96 h <i>Oncorhynchus mykiss</i> mg/L LC50 static 105: 96 h <i>Lepomis macrochirus</i> mg/L LC50 static		96: 48 h <i>Daphnia magna</i> mg/L EC50
2,2',2''-Nitrilotriethanol	216: 72 h <i>Desmodesmus subspicatus</i> mg/L EC50 169: 96 h <i>Desmodesmus subspicatus</i> mg/L EC50	10600-13000: 96 h <i>Pimephales promelas</i> mg/L LC50 flow-through 1000: 96 h <i>Pimephales promelas</i> mg/L LC50 static 450-1000: 96 h <i>Lepomis macrochirus</i> mg/L LC50 static		1386: 24 h <i>Daphnia magna</i> mg/L EC50
Neutralised boric acid	>28: 72 h <i>Selenastrum capricornutum</i> mg/L EC50	1020: 72 h <i>Carassius auratus</i> mg/L LC50 flow-through 627: 96 h <i>Oncorhynchus tshawytscha</i> mg/L LC50		115 - 153: 48 h <i>Daphnia magna</i> mg/L EC50
Dodecan-1-ol, ethoxylated		1.5: 96 h <i>Salmo salar</i> mg/L LC50		0.99: 48 h <i>Daphnia magna</i> mg/L EC50
3-Iodo-2-propynyl butylcarbamate	0.053: 72 h <i>Scenedesmus subspicatus</i> mg/L EC50	0.067: 96 h <i>Oncorhynchus mykiss</i> mg/L LC50		0.16: 48 h <i>Daphnia Magna</i> mg/L EC50

12.2. Persistence and degradability

No information available.

12.3. Bioaccumulative potential

Chemical Name	log Pow
Neutralised Dicyclohexylamine	3.5
Neutralised 1-Aminopropan-2-ol	-0.94
1-Aminopropan-2-ol	-0.94
Neutralised nonanoic acid	3.4
2,2',2''-Nitrilotriethanol	-2.53
Neutralised boric acid	-0.757
3-Iodo-2-propynyl butylcarbamate	2.8

12.4. Mobility in soil

No information available

12.5. Results of PBT and vPvB assessment

This preparation contains no substance considered to be persistent, bioaccumulating nor toxic (PBT). This preparation contains no substance considered to be very persistent nor very bioaccumulating (vPvB).

12.6. Other adverse effects

None known

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues / Unused Products	Dispose of as hazardous waste in compliance with local and national regulations
Contaminated packaging	Empty containers should be taken to an approved waste handling site for recycling or disposal. Observe all label precautions until container is cleaned, reconditioned or destroyed.
Other Data	According to the European Waste Catalogue, Waste Codes are not product specific, but application specific. Waste codes should be assigned by the user based on the application for which the product was used.

SECTION 14: TRANSPORT INFORMATION

14.1. UN-Number

UN3082

14.2. UN proper shipping name

ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (NEUTRALISED DICYCLOHEXYLAMINE, (Z)-OCTADEC-9-ENYLAMINE, ETHOXYLATED)

14.3. Transport hazard class

9

14.4. Packing group

III

14.5. Environmental Hazards

Yes

14.6. Special precautions for users

None

14.7. Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable

IMDG/IMO

UN No.	UN3082
Proper shipping name	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (NEUTRALISED DICYCLOHEXYLAMINE, (Z)-OCTADEC-9-ENYLAMINE, ETHOXYLATED)
Hazard class	9
Packing Group	III
EmS No.	F-A, S-F
Vessel Stowage Location Code	A
Marine Pollutant	This product contains a chemical which is listed as a marine pollutant according to IMDG/IMO
Description	UN3082, ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (NEUTRALISED DICYCLOHEXYLAMINE, (Z)-OCTADEC-9-ENYLAMINE, ETHOXYLATED), 9, III, Marine Pollutant

ADR/RID

UN No.	UN3082
Proper shipping name	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (NEUTRALISED DICYCLOHEXYLAMINE, (Z)-OCTADEC-9-ENYLAMINE, ETHOXYLATED)
Hazard class	9
Packing group	III
Classification Code	M6
ADR Hazard Id (Kemmler Number)	90
Environmental hazard	Yes
Description	UN3082, ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (NEUTRALISED DICYCLOHEXYLAMINE, (Z)-OCTADEC-9-ENYLAMINE, ETHOXYLATED), 9, III (E)

IATA

UN No.	UN3082
Proper shipping name	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (NEUTRALISED DICYCLOHEXYLAMINE, (Z)-OCTADEC-9-ENYLAMINE, ETHOXYLATED)
Hazard class	9
Packing group	III
ERG Code	9L
Description	UN3082, ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (NEUTRALISED DICYCLOHEXYLAMINE, (Z)-OCTADEC-9-ENYLAMINE, ETHOXYLATED), 9, III

ADN

UN/ID No	UN3082
Proper shipping name	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (NEUTRALISED DICYCLOHEXYLAMINE, (Z)-OCTADEC-9-ENYLAMINE, ETHOXYLATED)
Hazard class	9
Packing Group	III
Environmental hazard	Yes
Classification Code	M6
Hazard Labels	9
Equipment Requirements	PP
Description	UN3082, ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (NEUTRALISED DICYCLOHEXYLAMINE, (Z)-OCTADEC-9-ENYLAMINE, ETHOXYLATED), 9, III

SECTION 15: REGULATORY INFORMATION**15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture**

The Classification, Labeling and Packaging of Substances and Mixtures (CLP) Regulation (EC 1272/2008)
Registration, Evaluation, Authorization, and Restriction of Chemicals (REACH) Regulation (EC 1907/2006)

Statutory Instruments: Control of Substances Hazardous to Health Regulations 2002. Chemicals (Hazard Information and Packaging) Regulations 2009.

Acts of Parliament: The Health and Safety at Work etc. Act 1974. Environment Protection Act 1990.

Regulation on classification, labeling, of hazardous chemicals (2002 changing 2005). Appendix VI to Regulation on classification, labeling etc. of hazardous chemicals (2002 changing 2010), list of hazardous substances (as amended). Guidelines for submission and declaration of hazardous waste (2009). Transport of dangerous goods: ADR, RID, IMDG and IATA. Administrative norms for pollution of the atmosphere, 2009.

Workplace exposure limits (EH40)

WGK Classification

Severe hazard to water/Class 3

The highly refined base oil (Viscosity >20.5 cSt @40°C) contains one or more substance with the following CAS/EC numbers/REACH registration numbers:

Chemical Name	CAS-No	EC-No	REACH Registration Number
Lubricating oils (petroleum), C24-50, solvent-extd., dewaxed, hydrogenated	101316-72-7	309-877-7	01-2119489969-06-xxxx
Lubricating oils (petroleum), used, noncatalytically refined	101316-73-8	309-878-2	02-2119822310-56-xxxx
Distillates (petroleum), solvent-refined heavy paraffinic	64741-88-4	265-090-8	
Distillates (petroleum), solvent-refined light paraffinic	64741-89-5	265-091-3	01-2119487081-40-xxxx
Residual oils (petroleum), solvent deasphalted	64741-95-3	265-096-0	01-2119487081-40-xxxx
Distillates (petroleum), solvent-refined heavy naphthenic	64741-96-4	265-097-6	01-2119483621-38-xxxx
Distillates (petroleum), solvent-refined light naphthenic	64741-97-5	265-098-1	01-2119480374-36-xxxx
Residual oils (petroleum), solvent-refined	64742-01-4	265-101-6	01-2119488707-21-xxxx
Extracts (petroleum), residual oil solvent	64742-10-5	265-110-5	01-2119488175-30-xxxx
Distillates (petroleum), hydrotreated heavy naphthenic	64742-52-5	265-155-0	01-2119467170-45-xxxx
Distillates (petroleum), hydrotreated light naphthenic	64742-53-6	265-156-6	
Distillates (petroleum), hydrotreated heavy paraffinic	64742-54-7	265-157-1	01-2119484627-25-xxxx
Distillates (petroleum), hydrotreated light paraffinic	64742-55-8	265-158-7	01-2119487077-29-xxxx
Distillates (petroleum), solvent-dewaxed light paraffinic	64742-56-9	265-159-2	01-2119480132-48-xxxx
Residual oils (petroleum), hydrotreated	64742-57-0	265-160-8	01-2119489287-22-xxxx
Lubricating oils (petroleum), hydrotreated spent	64742-58-1	265-161-3	
Residual oils (petroleum), solvent-dewaxed	64742-62-7	265-166-0	01-2119480472-38-xxxx
Distillates (petroleum), solvent-dewaxed heavy, paraffinic	64742-65-0	265-169-7	01-2119471299-27-xxxx
Paraffin oils (petroleum), catalytic dewaxed heavy	64742-70-7	265-174-4	01-2119487080-42-xxxx
Paraffin oils (petroleum), catalytic dewaxed light	64742-71-8	265-176-5	01-2119485040-48-xxxx
Lubricating oils (petroleum), C>25, hydrotreated bright stock-based	72623-83-7	276-735-8	
Lubricating oils (petroleum), C20-50, hydrotreated neutral oil-based, high-viscosity	72623-85-9	276-736-3	01-2119555262-43-xxxx
Lubricating oils (petroleum), C15-30, hydrotreated neutral oil-based	72623-86-0	276-737-9	01-2119474878-16-xxxx
Lubricating oils (petroleum), C20-C50, hydrotreated neutral oil-based	72623-87-1	276-738-4	01-2119474889-13-xxxx
Lubricating oils	74869-22-0	278-012-2	
Paraffin oils	8012-95-1	232-384-2	
White mineral oil (petroleum)	8042-47-5	232-455-8	01-2119487078-27-xxxx
C18-C50 branched, cyclic and linear hydrocarbons – Distillates	848301-69-9	482-220-0	01-0000020163-82-xxxx

The highly refined, low viscosity mineral oils/hydrocarbons (Viscosity >7 - <20.5 cSt @40°C) contains one or more substance with the following CAS/EC numbers/REACH registration numbers:

Chemical Name	CAS-No	EC-No	REACH Registration Number
Distillates (petroleum), straight-run middle	64741-44-2	265-044-7	
Distillates (petroleum), heavy hydrocracked	64741-76-0	265-077-7	01-2119486951-26-xxxx
Distillates (petroleum), solvent-refined heavy paraffinic	64741-88-4	265-090-8	
Distillates (petroleum), solvent-refined light paraffinic	64741-89-5	265-091-3	01-2119487067-30-xxxx
Residual oils (petroleum), solvent deasphalted	64741-95-3	265-096-0	01-2119487081-40-xxxx
Distillates (petroleum), solvent-refined heavy naphthenic	64741-96-4	265-097-6	01-2119483621-38-xxxx
Residual oils (petroleum), solvent-refined	64742-01-4	265-101-6	01-2119488707-21-xxxx
Distillates (petroleum), hydrotreated middle	64742-46-7	265-148-2	01-2119459347-30-xxxx
Distillates (petroleum), hydrotreated middle	64742-46-7	934-956-3	01-2119827000-58-xxxx
Distillates (petroleum), hydrotreated light	64742-47-8	265-149-8	01-2119456620-43-xxxx
Distillates (petroleum), hydrotreated light naphthenic	64742-53-6	265-156-6	01-2119480375-34-xxxx
Distillates (petroleum), hydrotreated heavy paraffinic	64742-54-7	265-157-1	01-2119484627-25-xxxx
Distillates (petroleum), hydrotreated light paraffinic	64742-55-8	265-158-7	01-2119487077-29-xxxx
Distillates (petroleum), hydrotreated light paraffinic	64742-55-8	265-158-7	01-2119487077-29-xxxx
Distillates, petroleum, solvent-dewaxed light paraffinic	64742-56-9	265-159-2	01-2119480132-48-xxxx
Distillates (petroleum), solvent-dewaxed light paraffinic	64742-56-9	265-159-2	01-2119480132-48-xxxx
Residual oils (petroleum), hydrotreated	64742-57-0	265-160-8	01-2119489287-22-xxxx
Lubricating oils (petroleum), hydrotreated spent	64742-58-1	265-161-3	
Residual oils (petroleum), solvent-dewaxed	64742-62-7	265-166-0	01-2119480472-38-xxxx
Distillates (petroleum), solvent-dewaxed heavy, paraffinic	64742-65-0	265-169-7	01-2119471299-27-xxxx

Paraffin oils (petroleum), catalytic dewaxed light	64742-71-8	265-176-5	
Dec-1-ene, homopolymer, hydrogenated	68037-01-4	500-183-1	01-2119486452-34-xxxx
Lubricating oils (petroleum), C>25, hydrotreated bright stock-based	72623-83-7	276-735-8	
Lubricating oils (petroleum), C20-50, hydrotreated neutral oil-based, high-viscosity	72623-85-9	276-736-3	01-2119555262-43-xxxx
Lubricating oils (petroleum), C15-30, hydrotreated neutral oil-based	72623-86-0	276-737-9	01-2119474878-16-xxxx
Lubricating oils (petroleum), C20-C50, hydrotreated neutral oil-based	72623-87-1	276-738-4	01-2119474889-13-xxxx
Lubricating oils	74869-22-0	278-012-2	01-2119495601-36-xxxx
White mineral oil (petroleum)	8042-47-5	232-455-8	01-2119487078-27-xxxx
C18-C50 branched, cyclic and linear hydrocarbons – Distillates	848301-69-9	482-220-0	01-0000020163-82-xxxx
Hydrocarbons, C14-C19, isoalkanes, cyclics, <2% aromatics	NOT AVAILABLE	920-114-2	01-2119459347-30-xxxx

15.2. Chemical Safety Assessment

No information available

SECTION 16: OTHER INFORMATION

Key or legend to abbreviations and acronyms used in the safety data sheet

Repr.-Reproduction toxicity
 Asp. Tox. - Aspiration Toxicity
 Acute Tox. - Acute Toxicity
 Aquatic Acute - Acute Aquatic Toxicity
 Aquatic Chronic - Chronic Aquatic Toxicity
 Eye Dam. - Eye Damage
 Eye Irrit. - Eye Irritation
 Skin Corr. - Skin Corrosion
 Skin Irrit. - Skin Irritation
 Skin Sens. - Skin Sensitizer
 Resp. Sens. - Respiratory Sensitizer
 STOT SE - Specific target organ systemic toxicity (Single exposure)
 STOT RE - Specific target organ systemic toxicity (repeated exposure)
 VOC - Volatile organic compounds

Full text of H-Statements referred to under sections 2 and 3

• H224 - Extremely flammable liquid and vapor • H225 - Highly flammable liquid and vapor • H226 - Flammable liquid and vapor • H270 - May cause or intensify fire; oxidizer • H271 - May cause fire or explosion; strong oxidizer • H272 - May intensify fire; oxidizer • H290 - May be corrosive to metals • H300 - Fatal if swallowed • H301 - Toxic if swallowed • H302 - Harmful if swallowed • H304 - May be fatal if swallowed and enters airways • H310 - Fatal in contact with skin • H311 - Toxic in contact with skin • H312 - Harmful in contact with skin • H314 - Causes severe skin burns and eye damage • H315 - Causes skin irritation • H317 - May cause an allergic skin reaction • H318 - Causes serious eye damage • H319 - Causes serious eye irritation • H330 - Fatal if inhaled • H331 - Toxic if inhaled • H332 - Harmful if inhaled • H334 - May cause allergy or asthma symptoms or breathing difficulties if inhaled • H335 - May cause respiratory irritation • H336 - May cause drowsiness or dizziness • H340 - May cause genetic defects	• H341 - Suspected of causing genetic defects • H350 - May cause cancer • H351 - Suspected of causing cancer • H360 - May damage fertility or the unborn child • H361 - Suspected of damaging fertility or the unborn child • H362 - May cause harm to breast-fed children • H370 - Causes damage to organs • H371 - May cause damage to organs • H372 - Causes damage to organs through prolonged or repeated exposure • H373 - May cause damage to organs through prolonged or repeated exposure • H400 - Very toxic to aquatic life • H410 - Very toxic to aquatic life with long lasting effects • H411 - Toxic to aquatic life with long lasting effects • H412 - Harmful to aquatic life with long lasting effects • H413 - May cause long lasting harmful effects to aquatic life • H360Df - May damage the unborn child. Suspected of damaging fertility • H360D - May damage the unborn child • H360FD - May damage fertility. May damage the unborn child • H360F - May damage fertility • H361d - Suspected of damaging the unborn child • H361fd - Suspected of damaging fertility. Suspected of damaging the unborn child • H361f - Suspected of damaging fertility • EUH066 - Repeated exposure may cause skin dryness or cracking • EUH210 - Safety data sheet available on request • EUH208 - May produce an allergic reaction
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Exposure scenario

No information available

Revision Date:

07-14-2016

Revision Note**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.